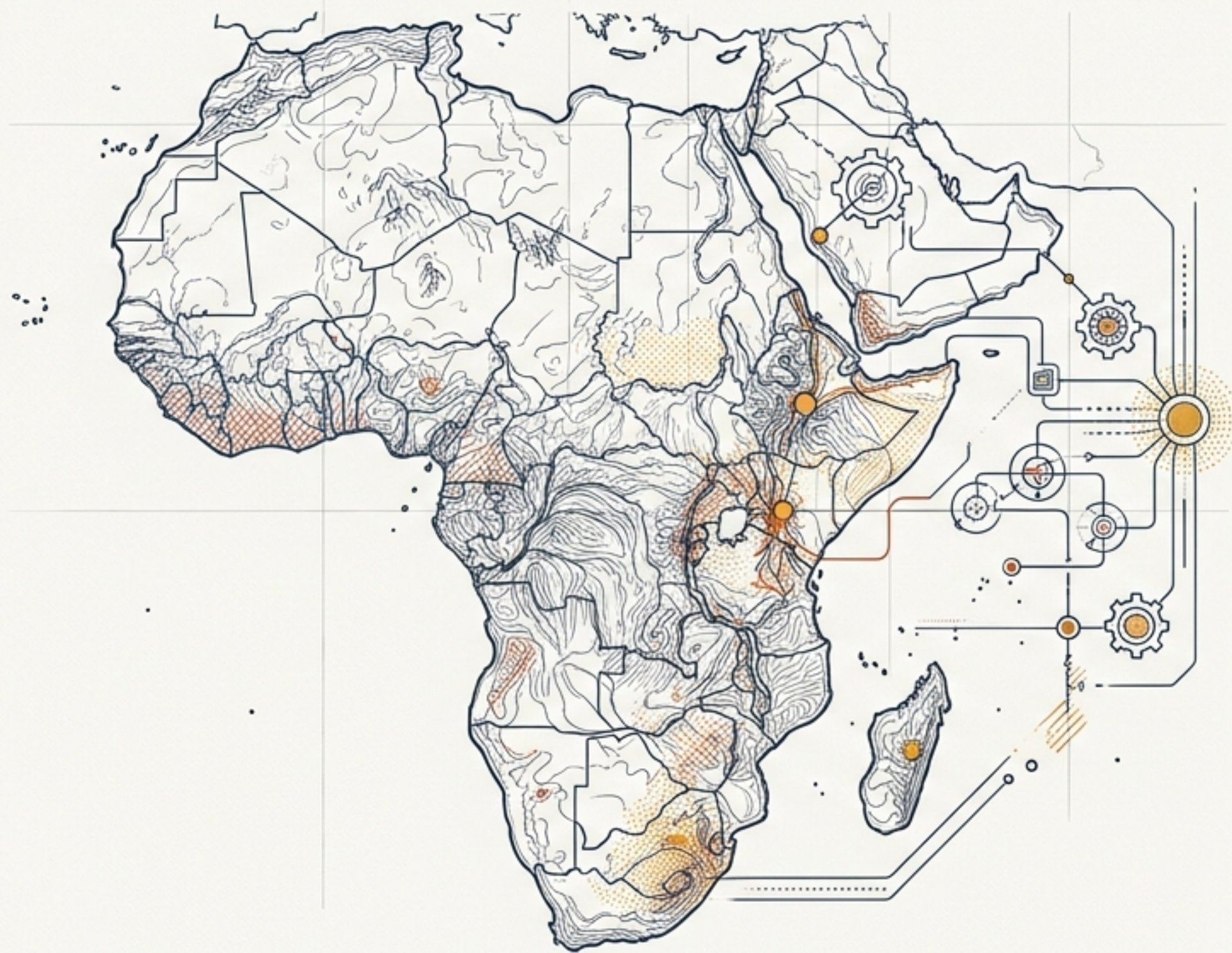
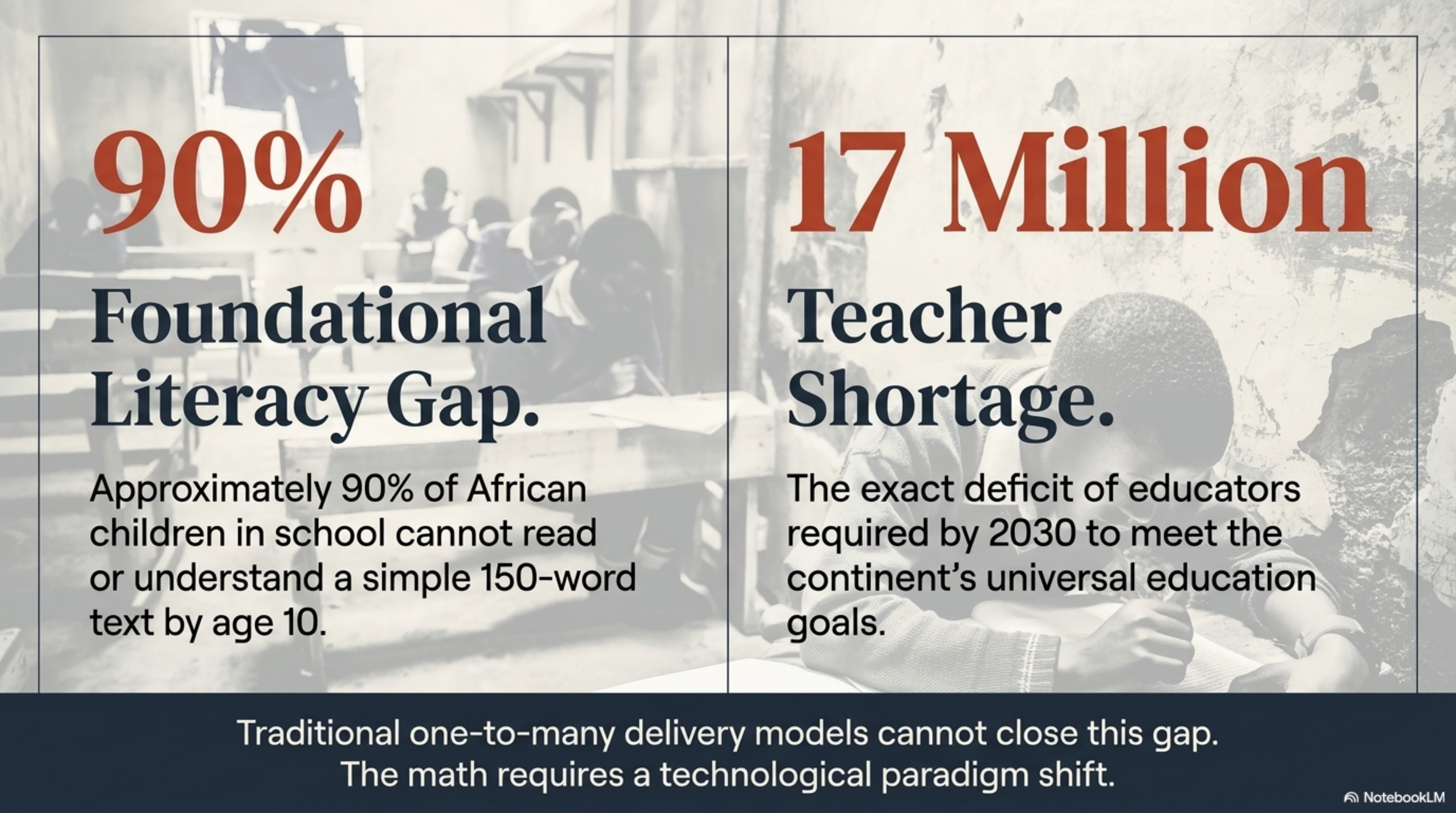


ALETU: The Adaptive Learning Engine for Africa

Scaling 1-to-1 Mastery through AI, Knowledge Graphs, and Value Density.





90%

Foundational Literacy Gap.

Approximately 90% of African children in school cannot read or understand a simple 150-word text by age 10.

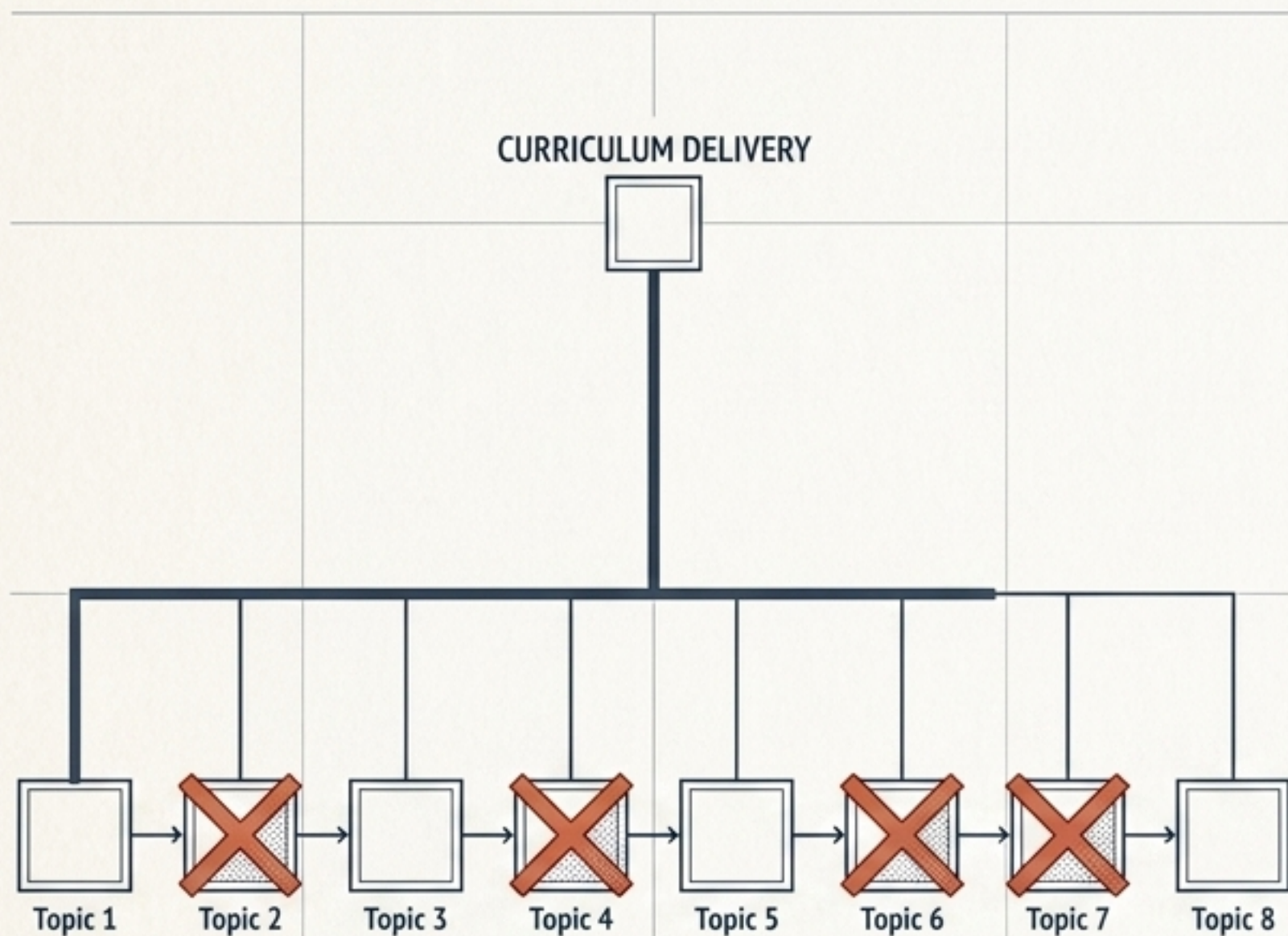
17 Million

Teacher Shortage.

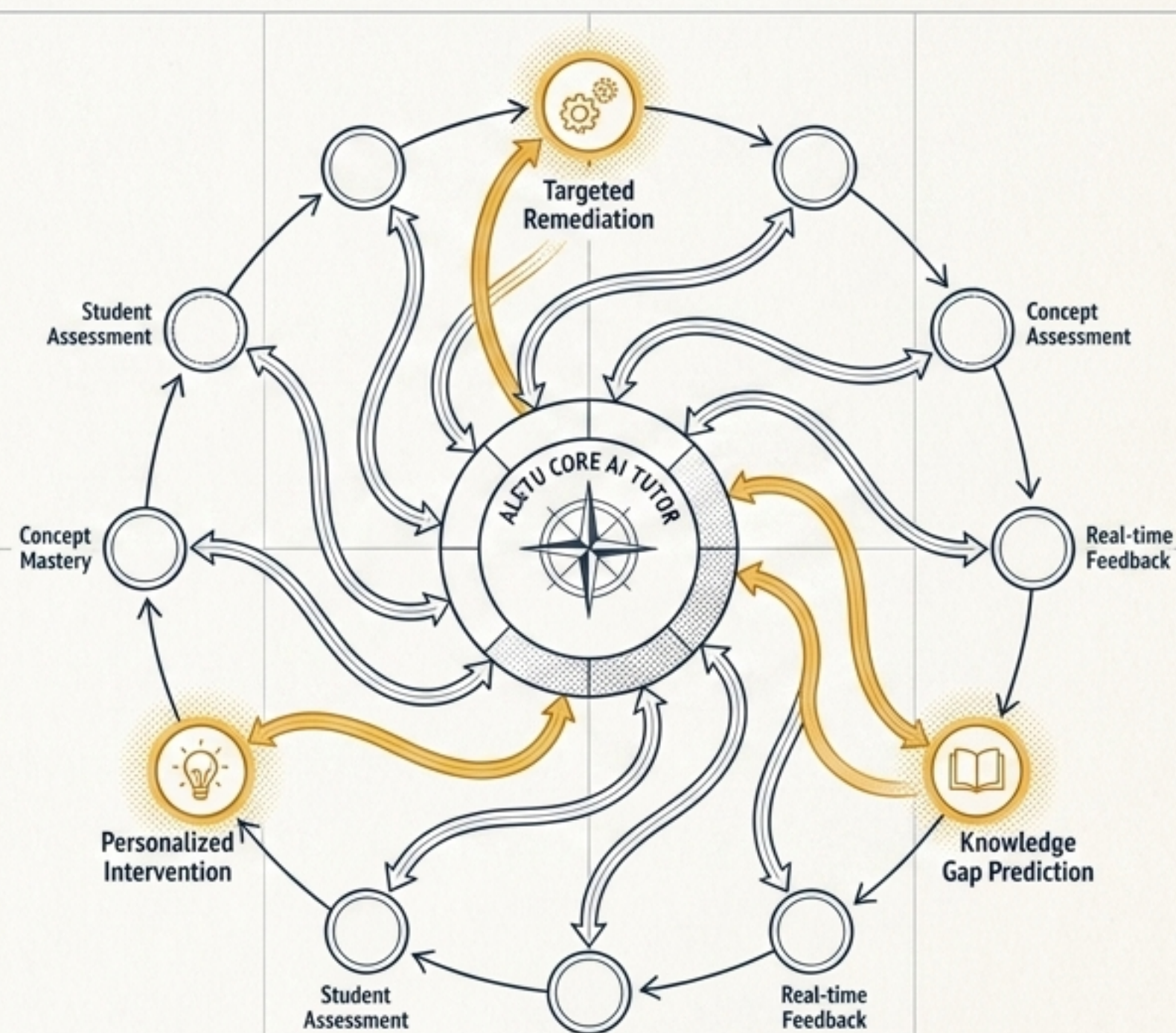
The exact deficit of educators required by 2030 to meet the continent's universal education goals.

Traditional one-to-many delivery models cannot close this gap.
The math requires a technological paradigm shift.

From Static Delivery to Dynamic Remediation



The traditional curriculum dictates a progression of topics regardless of comprehension. If you missed a concept at the beginning, you are left behind.



ALETU is a closed-loop AI tutor. It dynamically assesses state, predicts knowledge gaps, and intervenes with personalized remediation in real-time.

Context is the Catalyst: Grounding AI in Local Truth



LAYER 3: PERSONALIZED ASSESSMENT & CONTENT

LAYER 2: KNOWLEDGE GRAPH SYNTHESIS

LAYER 1: RAW DATA INGESTION

HIGH-FIDELITY INPUT

ALETU directly ingests the Ugandan NCDC K-12 syllabus.

STRUCTURAL INTEGRITY

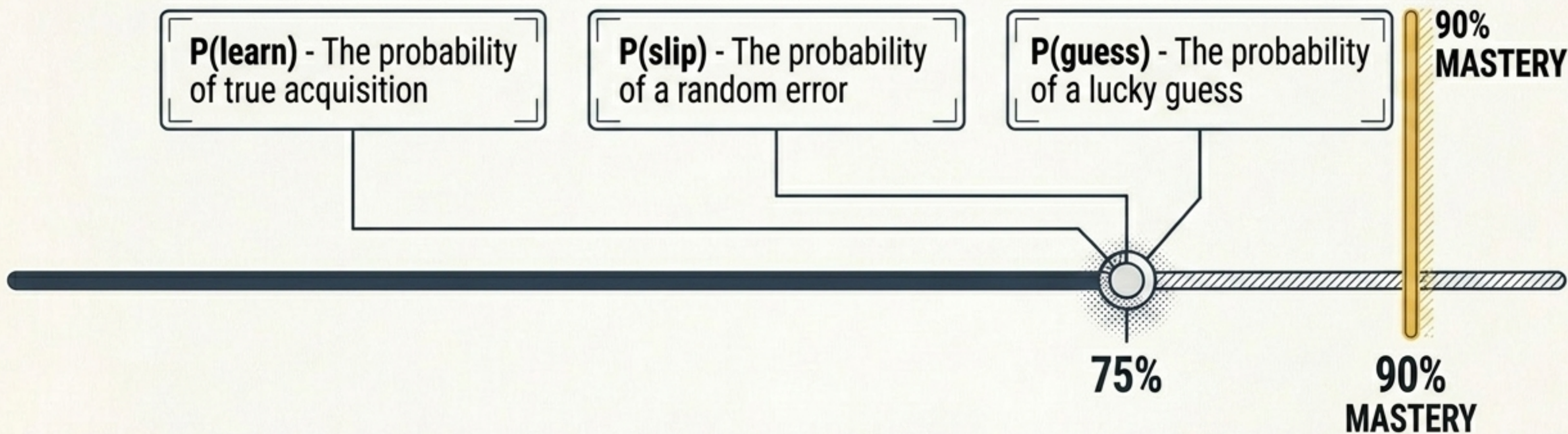
Raw data is converted into a proprietary Knowledge Graph.

CULTURAL RELEVANCE

Dedicated bias mitigation generates culturally aligned, hallucination-free content and assessments.

Bayesian Knowledge Tracing: The 90% Mastery Threshold

ALETU measures mastery, not just completion.

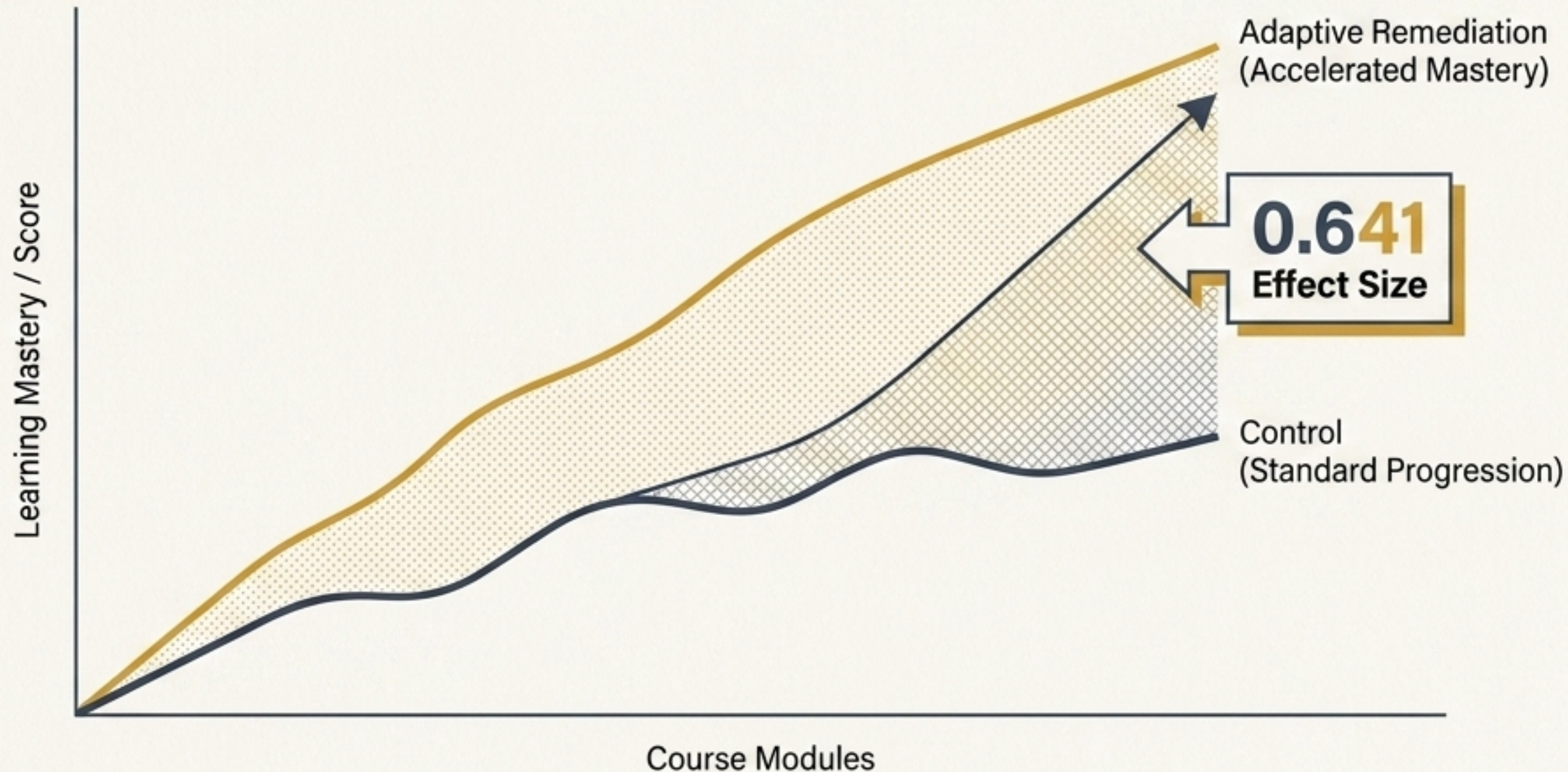


The algorithm continuously recalculates the probability a student understands a Knowledge Component. The system refuses to advance a learner until the algorithm calculates a $\geq 90\%$ probability of mastery.

Scientifically Proven Efficacy: The 0.641 Effect Size

Based on experimental implementation of adaptive learning in a Harvard MOOC, an engine that prioritizes remediation—serving items on topics with the least evidence of mastery—yields an extraordinary 0.641 learning gain effect size.

Adaptive paths do not just prevent students from falling behind; they actively accelerate mastery beyond traditional control groups.



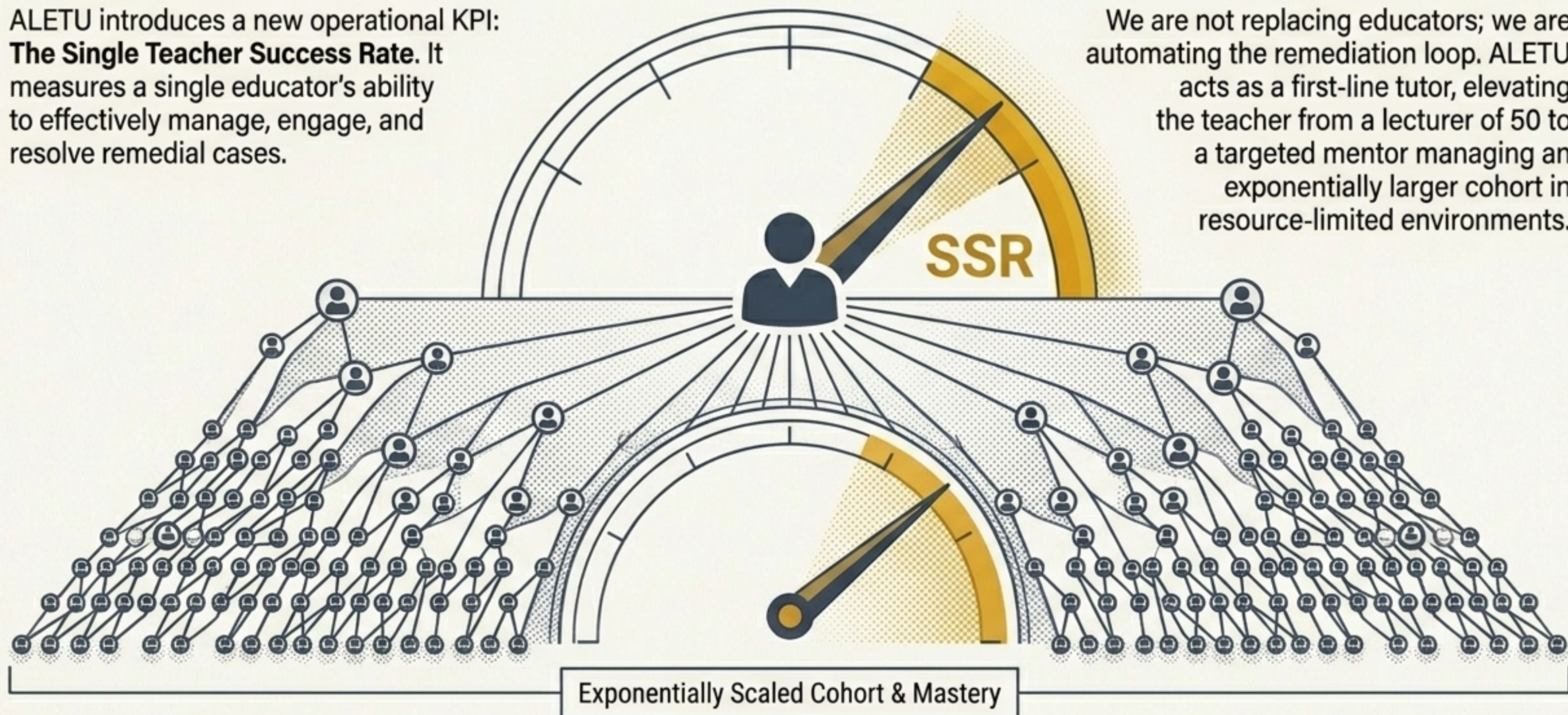
The Human Multiplier: Single Teacher Success Rate (SSR)

The Metric

ALETU introduces a new operational KPI: **The Single Teacher Success Rate**. It measures a single educator's ability to effectively manage, engage, and resolve remedial cases.

The Superpower

We are not replacing educators; we are automating the remediation loop. ALETU acts as a first-line tutor, elevating the teacher from a lecturer of 50 to a targeted mentor managing an exponentially larger cohort in resource-limited environments.



"AI companies tap labor budgets, not just software budgets. That's a 10x larger market opportunity hiding in plain sight."

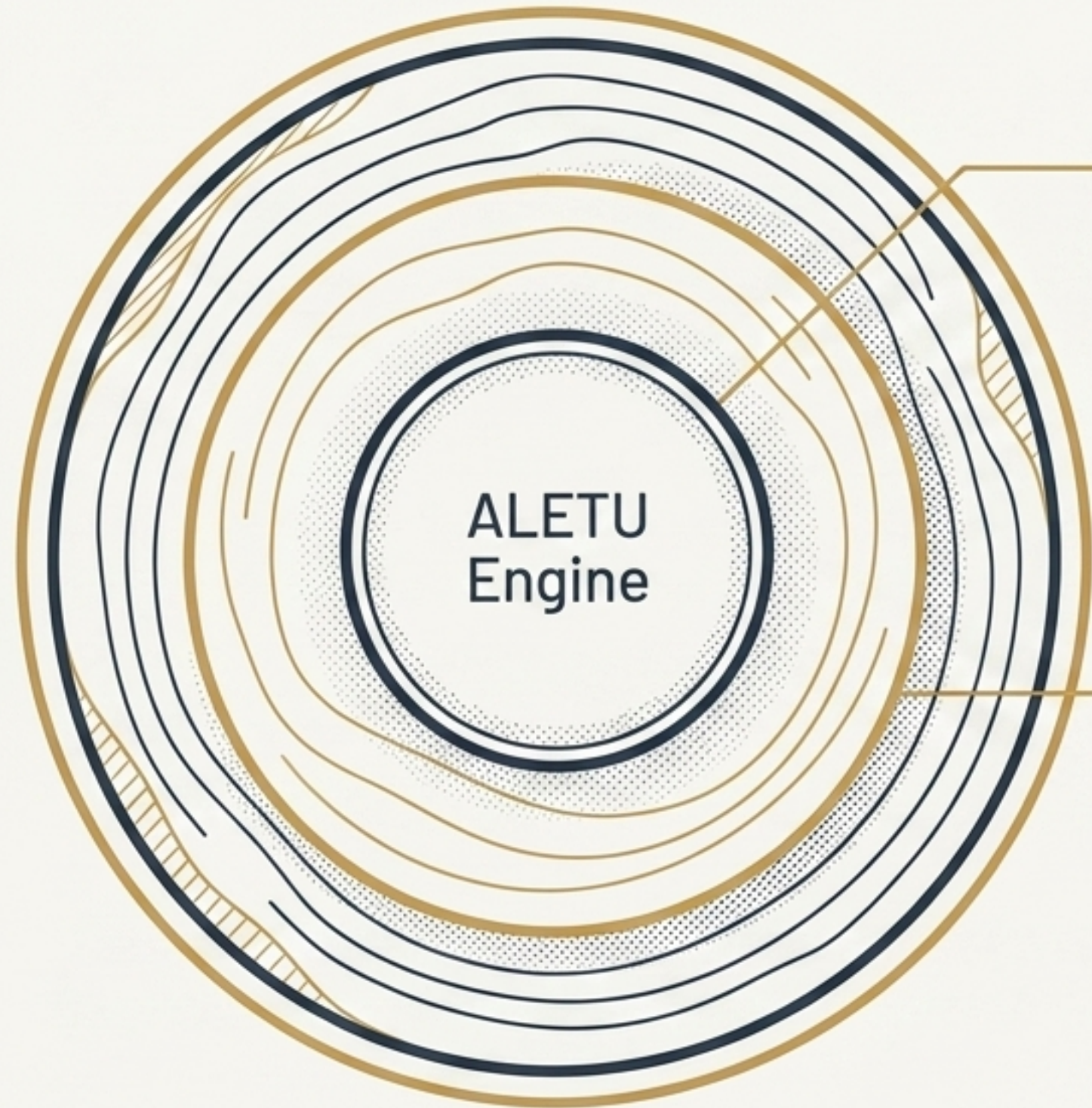
– Market Insight

The Economics of Value Density

Traditional EdTech SaaS	ALETU Agentic AI
Budget Source: School Software Budgets (Highly constrained).	Budget Source: Labor & Tutoring Budgets (A 10x larger market opportunity).
Margin Profile: High (70-85%), but fundamentally limited by low Average Revenue Per Account (ARPA).	Margin Profile: Moderate (50-60%), but drives astronomically higher absolute profit per customer due to high ARPA.
Core Value: Digitizing administration.	Core Value: Automating 1-to-1 cognitive labor.

By autonomously acting as a personalized remedial tutor, ALETU achieves unparalleled **Value Density**, fundamentally changing the unit economics per student.

Strategic Alignment: Riding the Continental Policy Wave



Uganda 2028 Mandate

Aligns directly with national targets: reaching 90% broadband coverage by geography, implementing the National Backbone Infrastructure (NBI), and digitizing the NCDC curriculum.

AUDA-NEPAD 2030 Mandate

Positioned as a Digital Public Infrastructure (DPI) aligned with the continental push for localized, offline-first, vendor-neutral educational ecosystems.

Product-Market Fit at Continental Scale



ALETU is the engine built to close the 90% learning gap.